

Lekhraj Kumar

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Summary

Senior Backend Software Engineer with 5+ years of experience designing and building scalable payment, banking, and SaaS platforms using Java and Spring Boot. Strong expertise in payment integrations, distributed systems, event-driven microservices, REST APIs, Kafka, and production-grade backend services. Experienced in designing reliable transaction processing systems, collaborating across cross-functional teams, and improving engineering productivity using AI coding tools including Claude Code, Cursor, and Kiro.

Education

National Institute of Technology Hamirpur

Dual Degree (Bachelor & Master of Technology), Electronics & Communication Engineering
Jul 2015 – Jun 2020

CGPA: 8.65/10

Skills

Programming: Java (Primary), Golang (working knowledge), JavaScript

Backend: Spring Boot, Spring Security, Hibernate/JPA, REST, GraphQL, WebSockets, OAuth2

Architecture: Microservices, Event-Driven Architecture, Domain-Driven Design, System Design(HLD/LLD), SaaS

Messaging: Kafka, RabbitMQ

Databases: MySQL, PostgreSQL, MongoDB, Redis

Cloud: Docker, Kubernetes, Helm, AWS, Tencent Cloud, GitHub Actions, Jenkins

Observability: Prometheus, ELK, Datadog, Centralized Logging, Monitoring & Alerting

Testing: JUnit, Mockito, Integration Testing, Contract Testing, TDD

Frontend: React.js, Mantine, Wicket

Engineering: Agile/Scrum, Code Reviews, On-call Support, RCA, Cross-functional Collaboration

Payments: Payment Gateway Integration, Merchant Integration, Authorization, Capture, Refund, Reversal, Settlement, Reconciliation, PCI DSS, Tokenization.

Professional Experience

CP Axta – Senior Software Engineer

Jun 2026–Present

- Designing and developing scalable backend services supporting payment, invoice, and transaction processing using Java 21/25, Spring Boot, PostgreSQL, Kafka, GraphQL, REST APIs, and CDC.
- Building event-driven backend services and asynchronous payment workflows using Kafka and CDC to enable reliable transaction processing and distributed data synchronization.
- Collaborating with Product, QA, DevOps, and Business teams to analyse requirements, identify technical risks, and deliver scalable backend payment solutions.
- Performing code reviews, mentoring engineers, and driving engineering best practices to deliver secure, maintainable, and production-ready backend services.
- Supporting production payment systems through incident investigation, root cause analysis, bug fixes, monitoring, and continuous reliability improvements.
- Contributing to platform reliability and engineering excellence by improving automated testing, CI/CD pipelines, dependency management, and build quality using modern development practices and AI-assisted coding tools.

AppDirect – Software Development Engineer II

Sep 2022–Jun 2026

- Designed and owned high-availability backend microservices supporting billing, checkout, notifications, and payment processing while maintaining p99 latency below 200ms and 99.9% availability.
- Built secure payment integrations with PayPal, Stripe, and Billpay using REST APIs, supporting billing, payment processing, merchant integration, and transaction workflows.
- Migrated legacy RabbitMQ-based event pipelines to Kafka, improving event throughput by 4x and enabling horizontal scalability, replayability, and fault-tolerant processing for billing and payment workflows handling millions of events per day.
- Contributed to decomposing a monolithic system into domain-aligned microservices using event-driven architecture, enabling independent scaling, faster deployments, and improved fault isolation.
- Practiced TDD with JUnit and Mockito, delivering comprehensive unit and integration tests, and built automated CI/CD pipelines using GitHub Actions.
- Participated in production support, on-call rotations, incident management, RCA, performance optimization, and long-term reliability improvements.

- Collaborated with Product Managers, QA, DevOps, and external partners to deliver multiple payment platform enhancements from design through production deployment.

Infosys – Senior Systems Engineer

Nov 2020–Sep 2022

- Developed backend services supporting banking transactions, payment workflows, and regulatory requirements for enterprise banking platforms.
- Migrated legacy Finacle scripts to modern Java-based microservices, improving maintainability, testability, deployment reliability, and fault isolation.
- Built and maintained EOD/BOD automation workflows to streamline time-sensitive banking operations, reducing manual effort and operational risk.
- Delivered secure backend services across CASA, Loans, and Deposits while ensuring transactional integrity, reconciliation accuracy, and system reliability.
- Investigated and resolved production issues, performed root cause analysis, and implemented fixes to stabilize live banking systems

Projects

SyncFlow – Real-Time Data Synchronization Platform

[Link](#)

Tech: Java 25, Spring Boot 3.5, Spring AI, PostgreSQL, MongoDB, Kafka, Redis, Docker, Kubernetes

- Architected a 17-module hexagonal platform using Spring Modulith and Virtual Threads, enforcing strict domain isolation with a framework-independent core.
- Engineered a Snapshot + CDC engine using Debezium and MongoDB Change Streams, processing 21K+ row changes/sec with retries, DLQs, and idempotent event processing.
- Built an AI Copilot leveraging OpenAI-compatible LLMs to generate pipeline mappings, diagnose failures, and optimize synchronization workflows.
- Designed a secure multi-tenant SaaS architecture with 19 RBAC permissions, tenant isolation, AES-256-GCM encryption, quota management, and immutable audit logs.
- Implemented cloud-native deployment using Kubernetes, Helm, ArgoCD, KEDA, Prometheus, Grafana, and LitmusChaos for autoscaling, observability, and resilience testing.
- Developed a zero-dependency Plugin SDK with isolated ClassLoaders, enabling third-party connector development without modifying the core platform.

Router Service

[Link](#)

Tech: Java 21, Spring Boot 3.5, Docker, WebSockets, Prometheus

- Designed and built a high-performance, enterprise-grade request router inspired by API gateways, focusing on low-latency routing, fault tolerance, and horizontal scalability.
- Implemented advanced routing using regex-based rules and multiple load-balancing strategies (Round Robin, Random, Least Connections).
- Added resilience patterns including circuit breakers, retries with exponential backoff, and token-bucket rate limiting to handle traffic spikes and downstream failures.
- Enabled real-time metrics streaming via WebSockets and built a monitoring dashboard for live traffic and health visibility.
- Optimized JVM memory, threading, and Docker multi-stage builds; validated performance with load testing up to 10,000+ RPS and 1M+ requests in stress scenarios.

Engineering Knowledge Assistant

[Link](#)

Tech: Java 25, Spring Boot 3.x, PostgreSQL/pgvector, Redis, Kafka, Spring AI

- Architected a 9-module hexagonal Spring Boot platform with clean architecture, domain-driven design, and plug-gable adapters for AI providers, persistence, messaging, caching, and external integrations.
- Designed an enterprise-grade RAG pipeline integrating GitHub, GitLab, Confluence, Jira, Swagger, PDFs, and web sources with semantic chunking, embeddings, hybrid retrieval (pgvector + PostgreSQL FTS), reranking, and context-aware conversational AI.
- Developed an event-driven ingestion framework using Kafka with asynchronous processing, idempotent consumers, dead-letter queues, and scalable document indexing.
- Implemented enterprise-grade security and observability with OAuth2/JWT authentication, Redis-backed rate limiting, OpenTelemetry tracing, Prometheus metrics, Jaeger, and structured logging.
- Designed a provider-agnostic AI layer supporting Anthropic, OpenAI, Ollama, Voyage AI, and Cohere through interchangeable adapters for LLMs, embeddings, and reranking.

Languages

- English (Professional)
- Hindi (Native)